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Source: *Army Ordnance*, JULY-AUGUST, 1924, Vol. 5, No. 25 (JULY-AUGUST, 1924), pp. 463-466

Published by: National Defense Industrial Association

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H. M. T. D.—A New Detonating Explosive*

By

C. A. TAYLOR and W. H. RINKENBACH

Hexamethylenetriperoxidediamine (H. M. T. D.) is a detonating compound, manufacture and use of which has been patented in Germany. It has recently been made the subject of an investigation by the explosives chemical laboratory of the Pittsburgh experiment station, United States Bureau of Mines, to determine whether it could be used as a substitute for mercury fulminate.

Legler[1] discovered H. M. T. D. while working on the slow combustion of ether. He obtained a solid, which, on treatment with ammonia, gave this compound, and noted some of its properties, including that of exploding on quick heating. Baeyer and Villiger[2] later carried out a more complete study of its preparation, properties and structure.

In 1912 von Girsewald announced[3] a new method of preparation, and in the same year patented the use of this substance as a primer and detonator[4], and also the method of manufacture[5]. In 1917, Leulier[6] published another method of preparation, and the results indicated that the compound had a constitution other than that previously assigned to it. However, von Girsewald[7] repeated Leulier's work, showed that the latter's new formula was incorrect, and later[8] suggested a somewhat different structure than that previously accepted.

Preparation of H. M. T. D.

Von Girsewald recommends dissolving 28 grams of hexamethylenetetramine (H. M. T.) and 42 grams of citric acid in 140 grams of 30 per cent hydrogen peroxide. He states in the patent specifications that the temperature of the solution rises, and a voluminous mass of H. M. T. D. separates out in the form of white, rhombic crystals. These are filtered off, washed with water and then with alcohol, and dried.

The directions given by von Girsewald are misleading. We found that the reaction is very rapid, and sufficient heat is developed to decompose the H. M. T. D., and the product first formed is oxidized without explosion, but with the evolution of considerable heat. Our experiments showed that the temperature should be kept below 30° C. during the reaction.

The modification of von Girsewald's method finally adopted by us was as follows: 56 grams of H. M. T. were dissolved in 185 grams of 30 per cent H_2O_2, or its equivalent of solution between 30 and 40 per cent. While cooling the mixture with a water-bath, 84 grams of powdered crystalline citric acid were then added, and the mixture stirred until all of the acid went into solution. The temperature was not allowed to rise above 30° C., and stirring was continued until the temperature began to decrease. The mixture was allowed to stand in a bath at room temperature for 6 to 15 hours, and was then filtered. The solid on the filter was washed several times with distilled water, and then rinsed with 95 per cent alcohol to facilitate drying. It dried quickly at room temperature. The undiluted filtrate was allowed to stand for several hours, and any additional precipitate was filtered off, washed, rinsed, dried, and added to the first portion obtained. From 68.5 to 71 per cent of the theoretical yield calculated from the weight of H. M. T. was obtained in each test.

From 56 grams of H. M. T., 280 grams of 30 per cent H_2O_2, and 84 grams of citric acid, a yield of 46.7 grams of H. M. T. D., and 285 grams of clear filtrate was obtained.

The filtrate deposited no more solid on standing for several days. It was a yellowish, slightly syrupy liquid, easily miscible with water. It contained free H_2O_2, and was strongly acid with citric acid. It was made up to two liters, and aliquot portions were distilled with KOH in order to determine the amount of ammonia present. It contained 13.409 grams of NH_3 and no unoxidized H. M. T. The reaction, then, may be as follows:

$$C_6H_{12}N_4 + 3\,H_2O_2 = C_6H_{12}N_2O_6 + 2\,NH_3$$

Grams	56.0	83.577	13.602
Grams found		46.7	13.409
Grams lost		36.9	

The H. M. T. D. not recovered may have decomposed without forming NH_3, according to the reactions:

$$C_6H_{12}N_2O_6 + 12\,O = N_2 + 6\,H_2O + 6\,CO_2$$

This shows that 22.8 per cent of the nitrogen originally introduced into the system as H. M. T. did not enter into the formation of H. M. T. D. Citric acid is used to neutralize the ammonia as formed and allow the formation of H. M. T. D.

Physical Properties of H. M. T. D.

Density—H. M. T. D. forms a white, powdery mass of minute rhombic crystals, which have no odor and are non-hygroscopic under ordinary conditions. Its true density at 20°/20° is 1.57, yet its apparent density after being placed in a tube and tapped was only 0.66,

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1Legler, *Berichte*, 18, 1885, p. 3344.
2Baeyer, Adolf, and Villiger, Victor, *Berichte*, 33, 1900, p. 2,486.
3von Girsewald, C., *Berichte*, 45, 1912, pp. 2,571-2,576.
4D. R. P. 274, 522, September 14, 1912.
5D. R. P. 263, 459, September 14, 1912.
6Leulier, *Jour. pharm. chim.* 15, 1917, pp. 222-229.
7von Girsewald, C., *Berichte*, 54, 1921, p. 390.
8von Girsewald, C., *Berichte*, 54, 1921, p. 393.

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and only 0.91 after being compressed in a detonator capsule under a pressure of 2,500 pounds per square inch.

Solubility—H. M. T. D. is only slightly soluble in most solvents at ordinary temperatures. The following table shows the solubility in some of the common solvents:

Solubility of H. M. T. D.

Solvent.	Grams per 100 grams of solvent at 22° C.
Water	0.01
Absolute alcohol	less than 0.01
Ether	0.017
Carbon bisulphide	less than 0.01
Carbon tetrachloride	0.013
Glacial acetic acid	0.14
Chloroform	0.64
Acetone	0.33

Sensitivity—H. M. T. D. detonates when struck a sharp blow, but a small quantity flashes like nitrocellulose when ignited, giving a flame several inches in length. Several samples were tested and found to require the impact produced by the fall of a 2,000-gram weight through a distance of 3 cm. in order to produce detonation, as compared with a fall of 0.25 cm. for mercury fulminate.

Various weights of H. M. T. D. were placed in No. 8 detonator capsules without reinforcing caps, subjected to a pressure of 1,000 pounds per square inch, and fired in the sand-test bomb. The weight of sand crushed in each case is given below, with comparisons for the same weight of charge of other detonating compounds:

Sand test with H. M. T. D., compared with other compounds.

Weight of charge, gram.	Sand crushed, grams				Sand crushed by same weight, grams	
	1	2	3	Average.	Mercury fulminate.	Cyanuric triazide.
0.050	3.3	3.3	3.2	3.3	—	2.6
0.100	6.3	6.8	6.8	6.6	—	4.8
0.150	10.2	9.4	10.2	10.0	—	8.0
0.200	17.7	14.4	15.5	15.9	3.8	12.2
0.300	25.0	22.1	23.9	23.7	7.8	21.5
0.500	44.0	42.4	41.2	42.5	16.5	44.2
0.750	66.4	64.9	65.2	65.5	26.5	65.5
1.000	86.3	87.1	86.6	86.7	36.6	78.6

These results show that H. M. T. D. is from 2½ to 3 times as strong as mercury fulminate, and slightly stronger than cyanuric triazide.

Initiator of detonation—In order to test the value of H. M. T. D. as an initiator of detonation, 0.40 grams of the explosive to be tested was weighed into a No. 8 detonator capsule, a desired charge of H. M. T. D. was added as a top layer, and pressed under 1,000 pounds pressure. The detonator so prepared was fired in the sand-test bomb. The sand crushed was weighed, and this weight indicated whether complete detonation of the high explosive had taken place. It was found that the use of a reinforcing cap was not necessary. Results are given in the following table:

Value of H. M. T. D. as an initiator of detonation.

Explosive.	Minimum detonating charge in grams.		
	Fulminate.	Hexamethylenetriperoxidediamine.	
		with reinforcing cap.	without reinforcing cap.
Trinitrotoluene	0.26	0.08	0.10
Picric acid	0.21	0.05	0.06
Tetryl	0.24	0.05	0.06
Ammonium picrate	0.8 to 0.9	0.30	0.30
Tetranitraniline	0.20	0.05	0.05
Guanidine picrate	0.30	0.13	0.15
Trinitroresorcin	0.20	0.08	0.10
Hexanitrodiphenylamine		0.05	0.05 --
Trinitrobenzaldehyde		0.08	0.10

These results show that only from one-fifth to one-third as much H. M. T. D. as mercury fulminate is necessary to cause the complete detonation of these high explosives. It does not become dead-pressed under 11,000 pounds pressure, while mercury fulminate only burns under 10,500 pounds.

Rate of Detonation—Weighed charges of H. M. T. D. were placed in No. 8 detonator shells, subjected to a pressure of 1,000 pounds per square inch, crimped to a black powder fuse, and embedded in the sticks of dynamite. The dynamites used were an ordinary 40 per cent nitroglycerin dynamite and a gelatin dynamite, which was insensitive, due to storage for over a year. Five-hundredths (0.05) of a gram of H. M. T. D. detonated each one. The rate of detonation of H. M. T. D., determined by the explosives testing laboratory, was 4,511 meters per second in a file 0.22-inch in diameter, and a loading density of 0.88.

Effect of heat—An explosive is worthless and may be dangerous if it is not stable under all conditions existing during its manufacture and use. Samples of H. M. T. D. were placed on watch glasses, heated in electric ovens, weighed, and examined at intervals. The following tabulation gives the loss in weight:

Per cent loss in weight by heating.

Temperature.	60°	75°	100°
Loss in 2 hours	0.10	0.25	3.25
Loss in 8 hours	0.35	0.60	29.60
Loss in 24 hours	0.50	1.30	67.95
Loss in 48 hours	0.50	2.25	

The sample heated at 60° showed no signs of decomposition.

The color of the sample heated at 75° was unchanged. It had a faint odor of methylamine, and had lost its dry character; it now appeared slightly moist and tended to "pack." Three 0.300-gram charges of the material that had been heated 48 hours at 75° C. were fired in the sand-test bomb. The weights of sand crushed were 22.6, 22.1 and 21.9 grams, respectively, an average of 22.2 grams, as compared with 23.7 grams for the unheated material.

The sample heated at 100° gradually softened and gave off an amine odor. The residue after heating at 100° C. for 24 hours consisted of a colorless fluid and needle-like crystals, which were soluble in water.

Effect of Storage—When first made, 2-gram charges crushed 15.9 grams of sand. This bulk material was allowed to stand on a desk in the laboratory for three months, when it crushed 14.1 grams. Some of the original detonators were tested at intervals of a month after standing in the open, and gave the following results:

	1	2	3	4	5	Ave.
1 month	11.7	13.5	11.4	13.6	11.8	12.4
2 months	7.7	9.6	9.4	5.3	7.7	8.0
3 months	2.1	2.5	2.1	4.7	2.8	2.8

Other detonators were kept in an air-tight container, and at the end of three months gave the following results:

15.2	14.9	15.6	15.4	14.8	15.2

Two samples were stored in sulphuric acid desiccators for three months. One was stored in the light and the other in the dark. The first lost 0.25 per cent in weight, and the latter 0.28 per cent. Detonators made from the material showed no loss in efficiency. Four samples of 3 grams each were sealed in weighing bottles and examined after three months storage. Two were exposed to the light and two were kept in the dark during this

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period. When opened all had the characteristic amine odor, but there was no apparent pressure developed, and 0.2-gram charges crushed an average of 15.0 grams of sand. Other samples were kept in small desiccators over water and crushed an average of 14.7 grams after three months storage.

The foregoing tests seem to indicate that equilibrium is established where the H. M. T. D. is stored in a closed vessel and that its efficiency is not materially decreased.

Hygroscopicity—Von Girsewald states that H. M. T. D. is not affected by water, even on long standing. In order to verify this statement, freshly prepared material was placed in a bottle, covered with distilled water, and allowed to stand for one month at room temperature. Five 0.300-gram charges were weighed into No. 8 detonator shells, and fired in the sand-test bomb. The remaining material was allowed to stand under water for another month, and the tests were repeated, giving the following results:

Time.	Sand crushed by 0.300 gm. charge of H. M. T. D. (grams).					Average.
Original	25.0	22.2	23.9	23.4	24.1	23.7
1 month	23.9	23.4	25.0	22.9	22.6	23.56
2 months	22.6	23.6	22.8	23.8	23.2	23.2

A quantity of H. M. T. D. was spread in a thin layer on a large watch glass, which was placed in a desiccator containing water, and the whole placed in an oven maintained at 30° C. for four weeks. At the end of each week some of the material was taken out, air dried, and five 0.300-gram charges were fired in the sand-test bomb. A slight amine odor was perceptible at the end of three weeks. Results are as follows:

Time. (Weeks)	Sand crushed by 0.300 gm. charge of H. M. T. D. (grams).					Average.
1	23.2	22.8	21.9	25.1	22.8	23.2
2	24.3	24.8	20.9	24.9	25.0	24.0
3	25.0	24.1	21.9	24.5	23.7	23.8
4	24.5	22.9	23.4	24.7	24.4	23.9

It is apparent from the foregoing that if decomposition takes place under these conditions, it does so to a very slight extent, and the residue is unimpaired as regards its explosive value.

That the presence of slight moisture will decrease this value was shown when, at the end of the fourth week, five shots were fired without previously drying the material. It was not moist, but showed a slight tendency to pack. The weight of sand crushed by 0.300-gram charges were 20.6, 22.2, 20.1, 19.6 and 18.3 grams respectively, an average of 20.2 grams.

Ignition temperature—When small quantities of H. M. T. D. were dropped on molten metal at 190° C., it decomposed, but detonated instantly at 200°. In three seconds, 0.05 gram ignited at 149° C., but decomposed without ignition at a lower temperature.

Chemical Properties of H. M. T. D.

H. M. T. D. is a neutral body, which does not appear to have any action upon metals or react with other substances to form addition products or molecular compounds.

Hydrolysis—Baeyer and Villiger, and Leulier, state that H. M. T. D. undergoes hydrolysis in acid solution. The writers of this paper found that hydrolysis will take place in acid, neutral, and alkaline solution, as shown by the following experiments:

Hydrolysis in neutral solution—Ten grams of the compound was added to 250 c. c. of distilled water, and the mixture boiled under a reflux condenser for half an hour. A tube led from the upper end of the condenser to a solution of ether, which was saturated with hydrogen chloride to catch any amines given off. The gas evolved contained oxygen, but no amines. The H. M. T. D. decomposed slowly in the water to a clear colorless liquid.

A portion of this liquor was analyzed for ammonia, and found to contain 95.3 per cent of that theoretically possible. Formaldehyde was also present. It is evident, then, that all, or practically all, of the nitrogen in this compound is transformed into ammonia on hydrolysis. This, together with the production of oxygen, would indicate the following as the equation of hydrolysis, as given by Leulier:

$$C_6H_{12}N_2O_6 + 6\,H_2O = 2\,NH_3 + 3\,C_2H_4\,(OH)_2 + 3\,O_2$$

The ethylene glycol then undergoes oxidation in the following manner:

$$C_2H_4\,(OH)_2 + 2\,O = H_2O + HCHO + HCOOH$$

Further reaction is then possible, the ammonia combining with the formic acid, and to a slight extent with the formaldehyde, to form H. M. T. All of these products were present.

Hydrolysis in acid solution.—Baeyer and Villiger, and Leulier, have previously studied the hydrolysis of H. M. T. D. with dilute sulphuric acid, and found the same reaction products as in neutral solution, so it was not considered necessary to repeat this work. However, it was found that this was also accomplished by organic acids at ordinary temperatures. Ten grams of H. M. T. D. was added to a solution of 42 grams of crystalline citric acid in 140 cc. of distilled water. It was found to be very slightly soluble, gradually going into solution with the evolution of oxygen, and gave off formaldehyde. It all decomposed in five days at room temperature. The solution so obtained was analyzed for ammonia and found to contain 85.4 per cent of that theoretically possible. It is evident, then, that the tendency to hydrolyze is increased by the presence of organic acids.

Hydrolysis in alkaline solution—It was found that when H. M. T. D. was treated with KOH solution, and allowed to stand at room temperature for a day, a strong odor of ammonia was given off, indicating that hydrolysis takes place in a manner similar to that in neutral or acid solution.

Oxidation of H. M. T. D.

If the reaction by which H. M. T. D. is produced is allowed to proceed unchecked, complete oxidation takes place, any of this substance at first formed being destroyed. In one test made by the writers, 28 grams of H. M. T. was dissolved in 140 grams of 30 per cent H_2O_2, and 42 grams of citric acid was added to the solution, which was stirred constantly. The temperature rose slowly to 34° C., and more rapidly to 45°, and apparently half of the usual yield of H. M. T. D. was precipitated. The temperature now rose rapidly to 80°, when violent boiling began, the temperature increasing to 111°. The precipitate of H. M. T. D. now decreased, and the appar-

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ent volume of the boiling mixture decreased. The temperature reached 121° C. before the precipitate was all consumed. The yellowish syrup remaining in the beaker was allowed to cool to room temperature, and weighed 84 grams.

This residue formed a thick, yellowish syrup of somewhat pungent odor. It was not easily miscible with water, but dissolved with shaking, and was made up to 1 liter for analysis.

The solution was strongly acid and gave a test for citric acid. A determination of ammonia by distillation with KOH showed 9.75 grams to be present in the total residue. Alkali does not decompose H. M., so a portion of the solution was heated to boiling with 3 per cent HCl, as Hartung[9] has shown that this hydrolyzes H. M. T. to formaldehyde, ammonia, and a little methylamine. The acid solution was cooled, made alkaline with KOH, and distilled for ammonia. No increase was found, showing no unchanged H. M. T. to be present in the solution.

Reduction of H. M. T. D.

Since H. M. T. D. is easily hydrolyzed in either neutral, acid, or alkaline media, it is to be expected that attempts at reduction would be met with only hydrolysis. A test with sodium amalgam gave no evidence of the formation of a product of reduction, but showed hydrolysis.

Other Reactions with H. M. T. D.

Leulier states that when H. M. T. D is treated with nitric acid, oxygen is given off. Concentrated sulphuric acid or bromine cause explosion. He also found that in the presence of dilute acid, this compound sets iodine free from potassium iodide, and so he used this method to determine its oxygen content.

H. M. T. D. and Potassium Chlorate Mixtures

Mixtures of these two substances were made up and tested in the sand-test bomb. A 30-70 mixture would give complete oxidation, this ratio giving the following results:

Sand test of 30-70 H. M. T. D.—$KClO_3$ mixture.

Charge of mixture,	Sand crushed, grams.				Sand crushed by equal change
gram.	1	2	3	Average.	H. M. T. D., grams.
0.200	10.5	11.0	9.5	10.3	15.9
0.300	18.2	16.7	16.3	17.1	23.7
0.500	29.4	27.7	27.5	28.2	42.5
0.750	46.4	47.7	47.0	47.1	65.5
1.000	62.3	63.6	61.2	62.3	86.7
1.500	93.8	95.2	—	94.5	—

These results are only from 65 to 72 per cent as high as those given by equal weights of straight H. M. T. D.

Other mixtures were made up in an effort to find a composition that would have a greater crushing effect than straight H. M. T. D., and the results obtained were:

H. M. T. D.	$KClO_3$	Sand crushed by 0.75 gm. charge, grams.		
per cent.	per cent.	1	2	Average.
30.0	70.0	46.4	47.7	47.1
50.0	50.0	53.3	50.9	52.1
70.0	30.0	61.7	62.9	62.3
100.0	0.0	67.2	65.6	66.4

[9]Hartung.—*Jour. fur prakt. Chemie*, 2, 46 (year ?) p. 16.

From this it is apparent that no mixture of these two substances will be more powerful than the pure H. M. T. D. Tests made on the small impact machine showed that these mixtures had the same sensitivity to impact as H. M. T. D. alone.

Physiological Properties of H. M. T. D.

H. M. T. D. is not an irritant, and does not cause dermatitis. It produces no other effect than violent sneezing when small amounts are inhaled. No apparent injurious effects were noted when it was fed to rats.

Composition of H. M. T. D.

Baeyer and Villiger, and von Girsewald, made a number of analyses of H. M. T. D., and agreed on the empirical formula $C_6H_{12}N_2O_6$, which the two former investigators checked by a molecular weight determination. Leulier announced this to be $C_4H_{11}NO_4$, but von Girsewald and Siegens[10] repeated his work, and showed his determination of the nitrogen content to be at fault, obtaining results which again confirmed the constitution as being $C_6H_{12}N_2O_6$.

Structure of H. M. T. D.

Baeyer and Villiger gave the structure the form (1)—

$$\begin{array}{c} / \ CH_2 - O - O - CH_2 \ \backslash \\ N - CH_2 - O - O \ CH_2 - N \\ \backslash CH_2 - O - O - CH_2 \ / \end{array} \qquad (1)$$

and this is the one commonly accepted today; but von Girsewald and Siegens[11] have since suggested the form (2)—

$$\begin{array}{l} O - CH_2 \ \backslash \qquad\qquad\qquad\qquad / \ CH_2 - O \\ | \qquad N - CH_2 - O - O - CH_2 - N \qquad | \\ O - CH_2 \ / \qquad\qquad\qquad\qquad \backslash CH_2 - O \end{array} \qquad (2)$$

This does not differ essentially from the form commonly accepted, and either formula agrees well with the chemical behavior of the compound.

Summary

A study of the preparation and properties of hexamethylenetriperoxidediamine (H. M. T. D.) has shown the following:

1. That the patent directions must be modified in regard to temperature.
2. That the amount of hydrogen peroxide required in preparation is considerably less than specified by the patentee.
3. That the compound is more powerful and efficient as a detonating agent and considerably less sensitive to impact than mercury fulminate.
4. That H. M. T. D. is stable at temperatures up to at least 60° C.; it is not affected by storage under water; but it is slowly affected when subjected to high humidity at maximum summer temperature, and very slowly under ordinary storage, but it will not explode when moist. It is practically non-hygroscopic.
5. That it is less dangerous than the azides to prepare and handle, and is not poisonous.

[10]von Girsewald, C., and Siegens, H., *Berichte*, 54B (1921), p. 490.

[11]Work cited, p. 493.

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